

ENTERPRISE ARCHITECTURE (EA) AND DATABASE PROGRAMMING

LECTURE 2

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CONTENTS:

- Enterprise Architecture
- The Architecture Process
- Drivers for Enterprise Architecture
- Enterprise Architecture and Other Governance Instruments

ENTERPRISE ARCHITECTURE

To create an integrated perspective of an enterprise, we need techniques for describing architectures in a coherent way and **communicating** these with **all relevant stakeholders**. Different types of stakeholders will have their own viewpoints on the architecture.

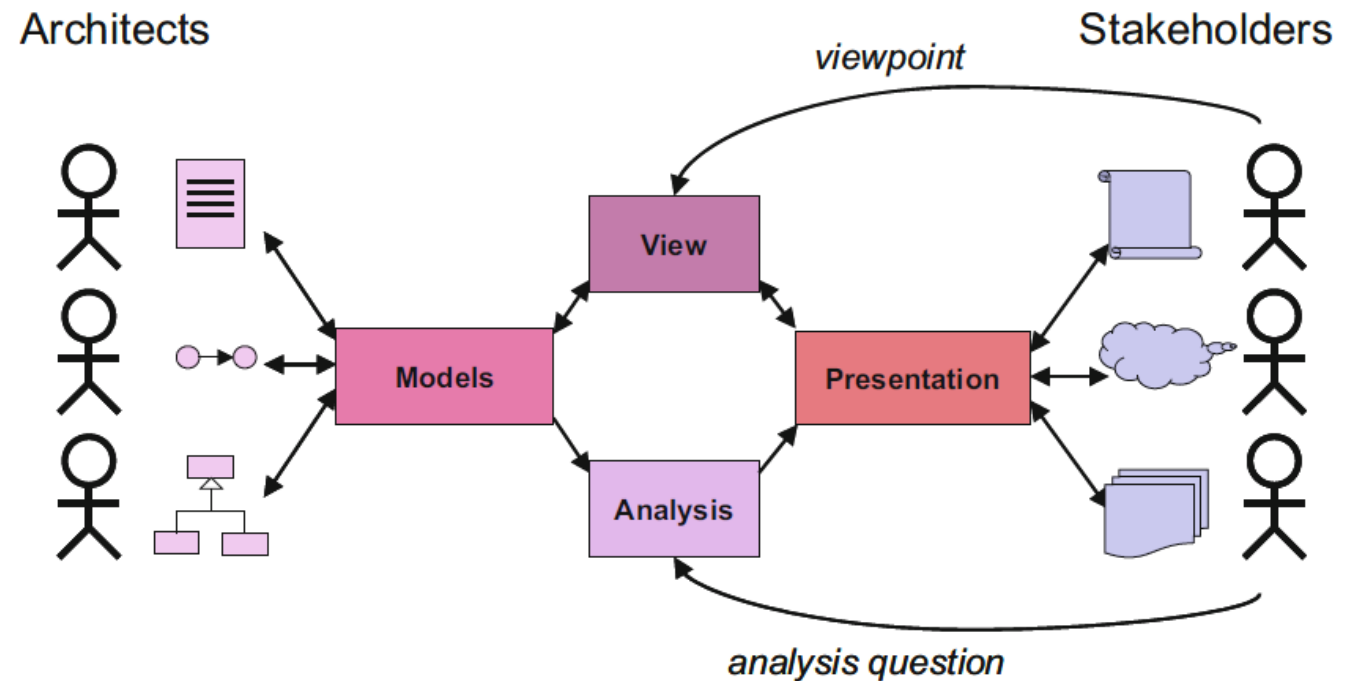


Fig. 1.1 Communicating about architecture

THE ARCHITECTURE PROCESS

The architecture process consists of the usual steps that take an initial idea through design and implementation phases to an operational system, and finally changing or replacing this system, closing the loop. In all of the phases of the architecture process, clear communication with and between stakeholders is indispensable.

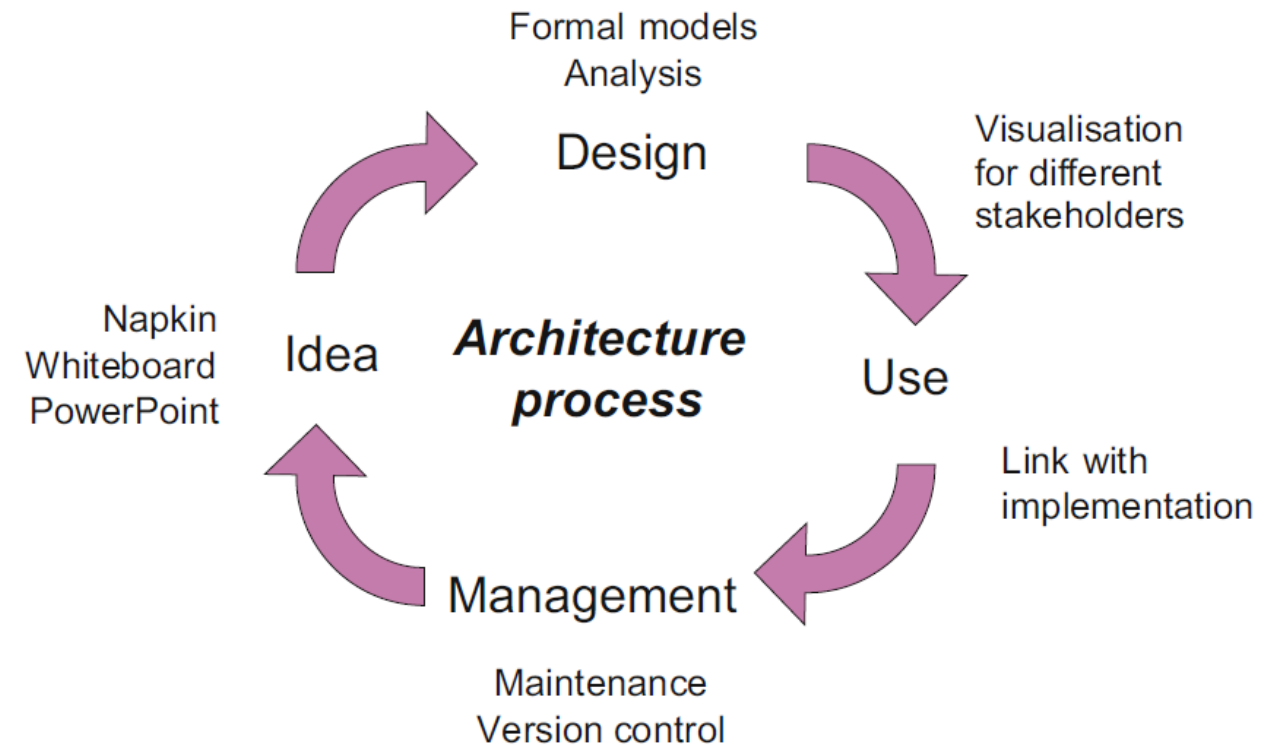


Fig. 1.2 The architecture description life cycle

DRIVERS FOR ENTERPRISE ARCHITECTURE

Drivers are the events, conditions or circumstances that **motivate an enterprise** to change some aspect of its goals and consequently its objectives, and potentially business activities or technical solutions.

There are many different drivers, including:

- **Legal or Regulatory** - for example, a new law governing Credit Card Usage
- **Cultural or Social** - for example acceptance of a product in a new market
- **Financial and Economic** - for example, reduction in profit due to changes of an exchange rate for borrowed foreign capital
- **Organizational or Cultural** - for example, high staff turn-over in a particular division or business unit

DRIVERS FOR ENTERPRISE ARCHITECTURE

Drivers:

- **Internal drivers** to execute effectively an organisation's strategy and optimise its operations (such as financial or budget pressures)
- **External drivers** (external pressures) that push organisations towards adopting enterprise architecture practice (such as regulatory obligations or forces from competitors)

They can be top-down, in the form of strategic drivers such as regulatory compliance and rule changes, or bottom-up in the form of tactical drivers such as technology changes or opportunities that have been discovered by audits, performance assessment, surveys or other mechanisms.

DRIVERS FOR ENTERPRISE ARCHITECTURE: INTERNAL DRIVERS

The well-known **strategic alignment model** of Henderson and Venkatraman (1993) distinguishes between the aspects of business strategy and organisational infrastructure on the one hand and IT strategy and IT infrastructure on the other hand (Fig. 1.3). The model provides **four dominant perspectives** that are used to tackle the alignment between these aspects. In any of these perspectives, an enterprise architecture can be a valuable help in executing the business or IT strategy.

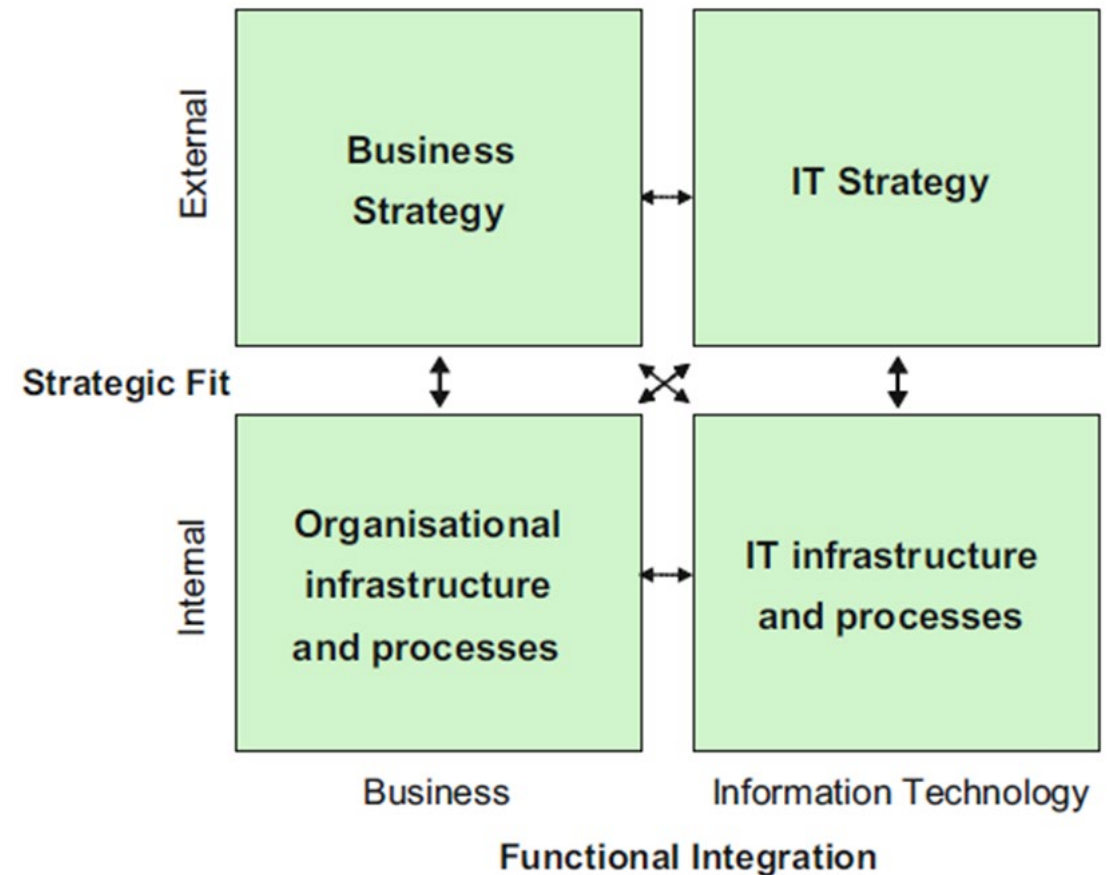


Fig. 1.3 Strategic alignment model (Henderson and Venkatraman 1993)

DRIVERS FOR ENTERPRISE ARCHITECTURE: INTERNAL DRIVERS

Nadler et al. (1992) identify four relevant alignment components:

- ❑ work
- ❑ people
- ❑ the formal organisation
- ❑ the informal organisation

DRIVERS FOR ENTERPRISE ARCHITECTURE: INTERNAL DRIVERS

Labovitz and Rosansky (1997) emphasise the horizontal and vertical alignment dimensions of an organisation.

- **Vertical alignment** describes the relation between the top strategy and the people at the bottom
- **Horizontal alignment** describes the relation between internal processes and external customers.

DRIVERS FOR ENTERPRISE ARCHITECTURE: INTERNAL DRIVERS

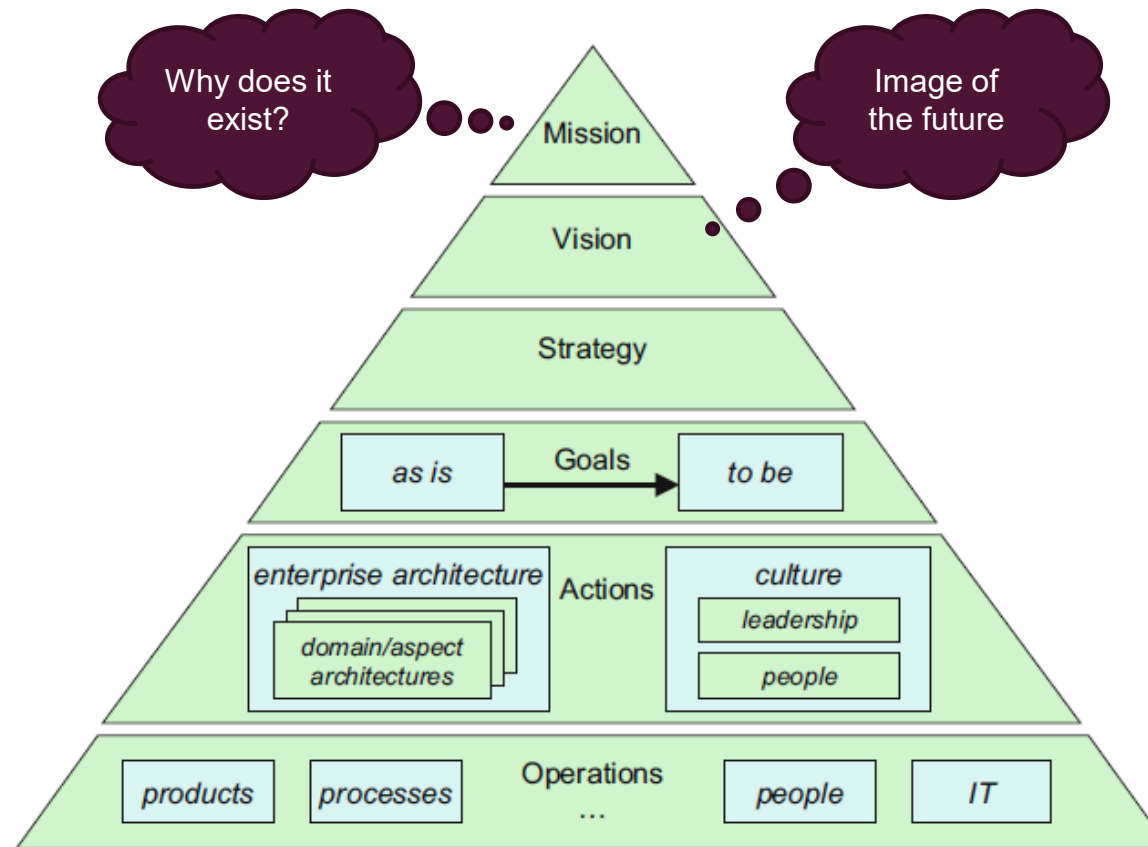


Fig. 1.4 Enterprise architecture as a management instrument

DRIVERS FOR ENTERPRISE ARCHITECTURE: INTERNAL DRIVERS

Moreover, architecture is a **strategic instrument** in guiding an organization through a planned course of development. As Ross et al. (2006) show with numerous case studies, successful enterprises employ an **'operating model'** with clear choices on the levels of integration and standardisation of business processes across the enterprise (Fig. 1.5). This operating model should fit both their area of business and their stage of development.

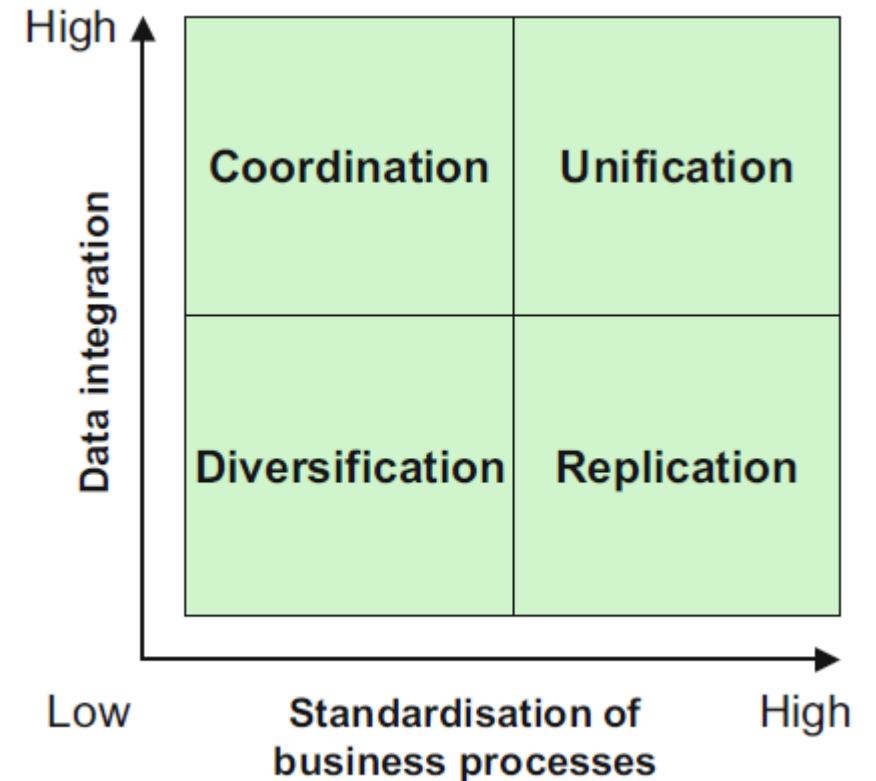


Fig. 1.5 Operating model (Ross et al. 2006)

DRIVERS FOR ENTERPRISE ARCHITECTURE: EXTERNAL DRIVERS

Example

- In the USA, the Clinger–Cohen Act (1996), also known as the Information Technology Management Reform Act, demands that every government agency must have an IT architecture, which is defined as: ‘an integrated framework for evolving or maintaining existing information technology and acquiring new information technology to achieve the agency’s strategic goals and information resources management goals’.

DRIVERS FOR ENTERPRISE ARCHITECTURE

Discussion

According to your business (project) describe **what are the internal and external drivers?**

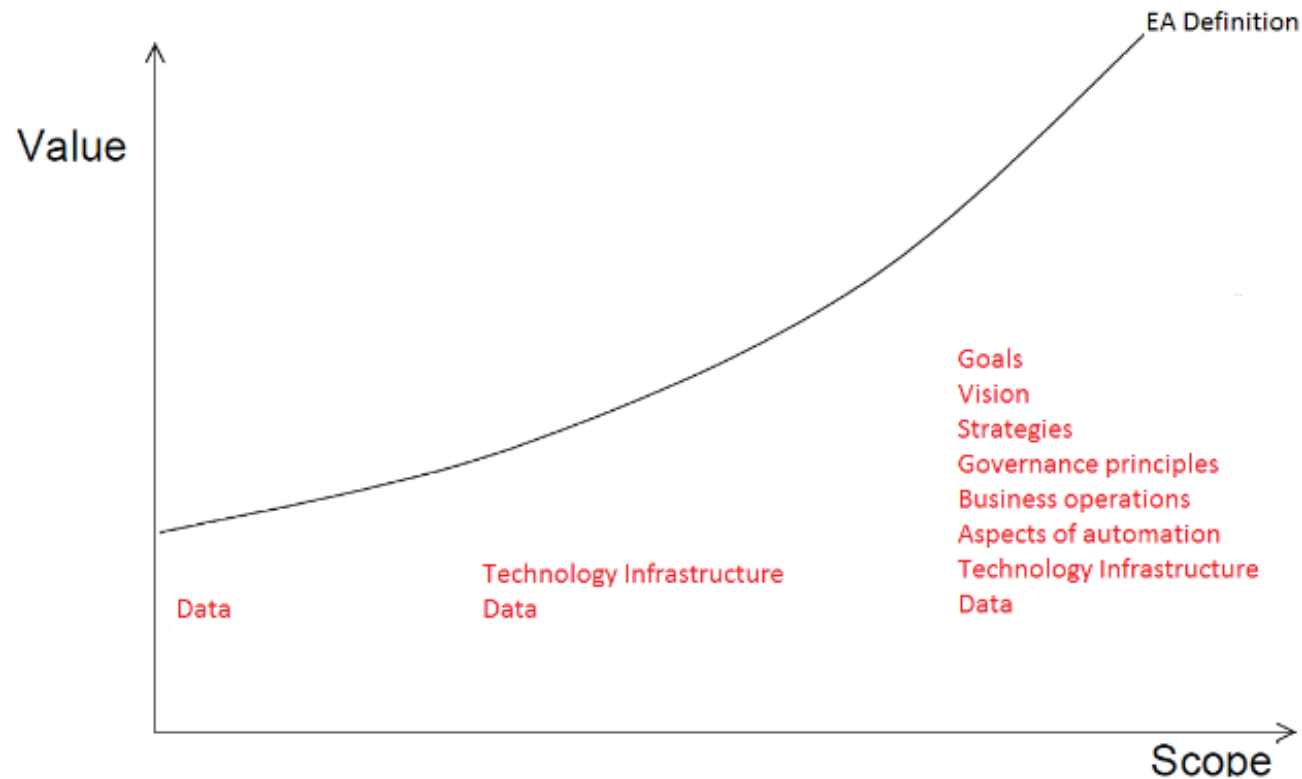


ENTERPRISE ARCHITECTURE: SUMMARY

- **Architecture** is the art and science of designing complex structures.
- **Enterprise architecture**, more specifically, is defined as a coherent whole of principles, methods, and models that are used in the design and realisation of an enterprise's organisational structure, business processes, information systems, and infrastructure.
- **Architecture models, views, presentations, and analyses** all help to bridge the 'communication gap' between architects and stakeholders.
- **Architecture** is an indispensable instrument in controlling the complexity of the enterprise and its processes and systems.
- On the one hand, we see **internal drivers** for using an architectural approach, related to the strategy execution of an organisation. Better alignment between business and IT leads to lower cost, high equality, better time-to-market, and greater customer satisfaction.
- On the other hand, **external drivers** from regulatory authorities and other pressures necessitate companies to have a thorough insight into their structure and operations.
- **All of these drivers** make a clear case for the use of enterprise architecture.

ENTERPRISE ARCHITECTURE: SUMMARY

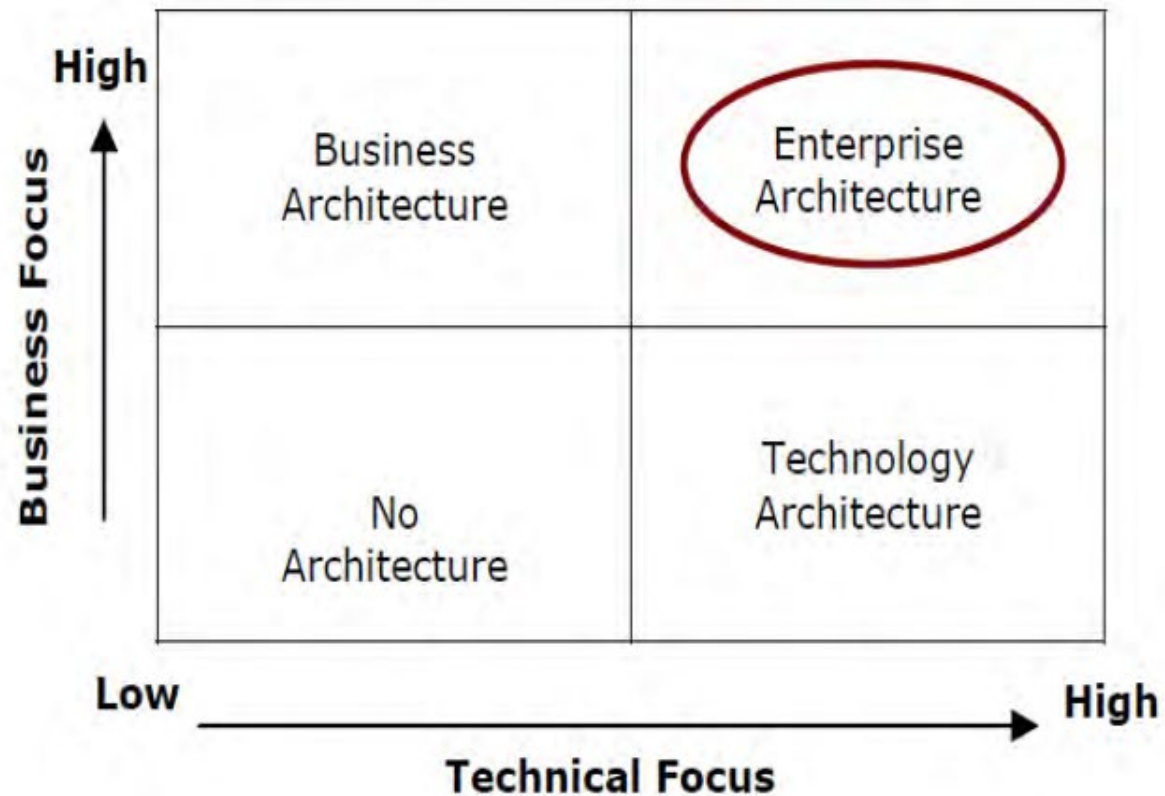
The Increasing Value of Enterprise Architecture



(Bredemeyer Consulting, 2003: 1)

ENTERPRISE ARCHITECTURE: SUMMARY

The Focus of Enterprise Architecture



ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

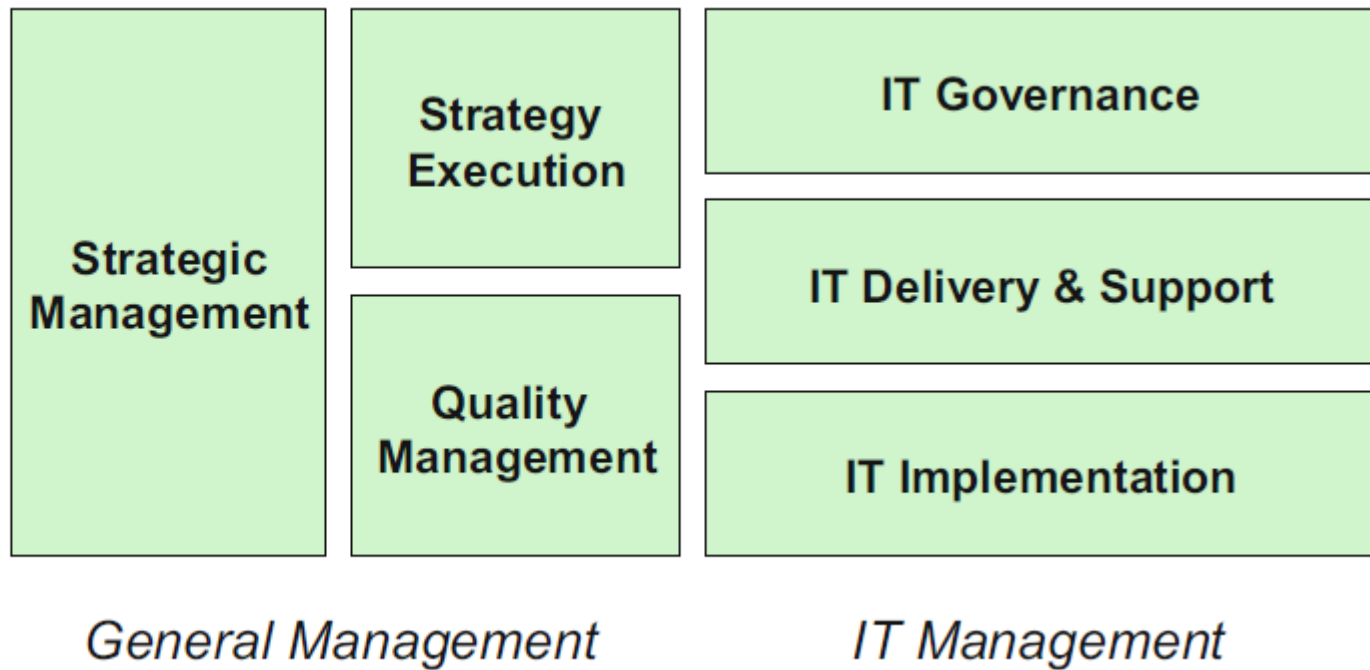
How does EA fit in with other established management practices and instruments?

How does EA position within the context of corporate and IT governance?



ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

Fig. 2.1 Management areas relevant to enterprise architecture



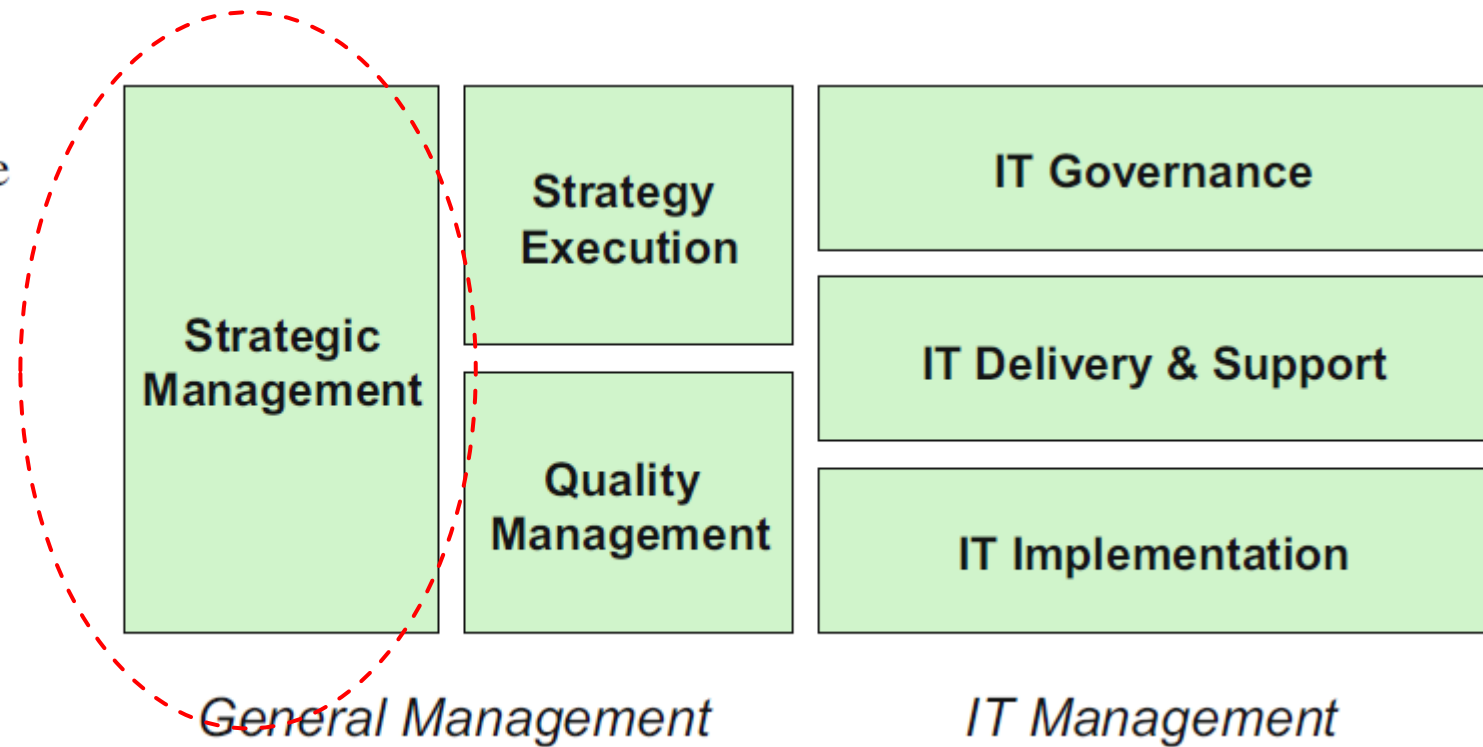
ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

There is a relationship between enterprise architecture with some well-known management practices in each of these areas, which can be shown as the position and role of enterprise architecture in a management context:

- ❑ **Strategic management:** Balanced Scorecard
- ❑ **Business model development:** Business Model Canvas
- ❑ **Business architecture:** BIZBOK® Guide and O-BA
- ❑ **Quality management:** EFQM and ISO 9001
- ❑ **IT governance:** COBIT
- ❑ **IT delivery and support:** ITIL
- ❑ **IT implementation:** CMM and CMMI

ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

Fig. 2.1 Management areas relevant to enterprise architecture



ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

Strategic Management

Kaplan and Norton (1992) introduced the **Balanced Scorecard (BSC)** as a management system that helps an enterprise to clarify and implement its vision and strategy. The BSC therefore suggests viewing an enterprise from **four perspectives**.

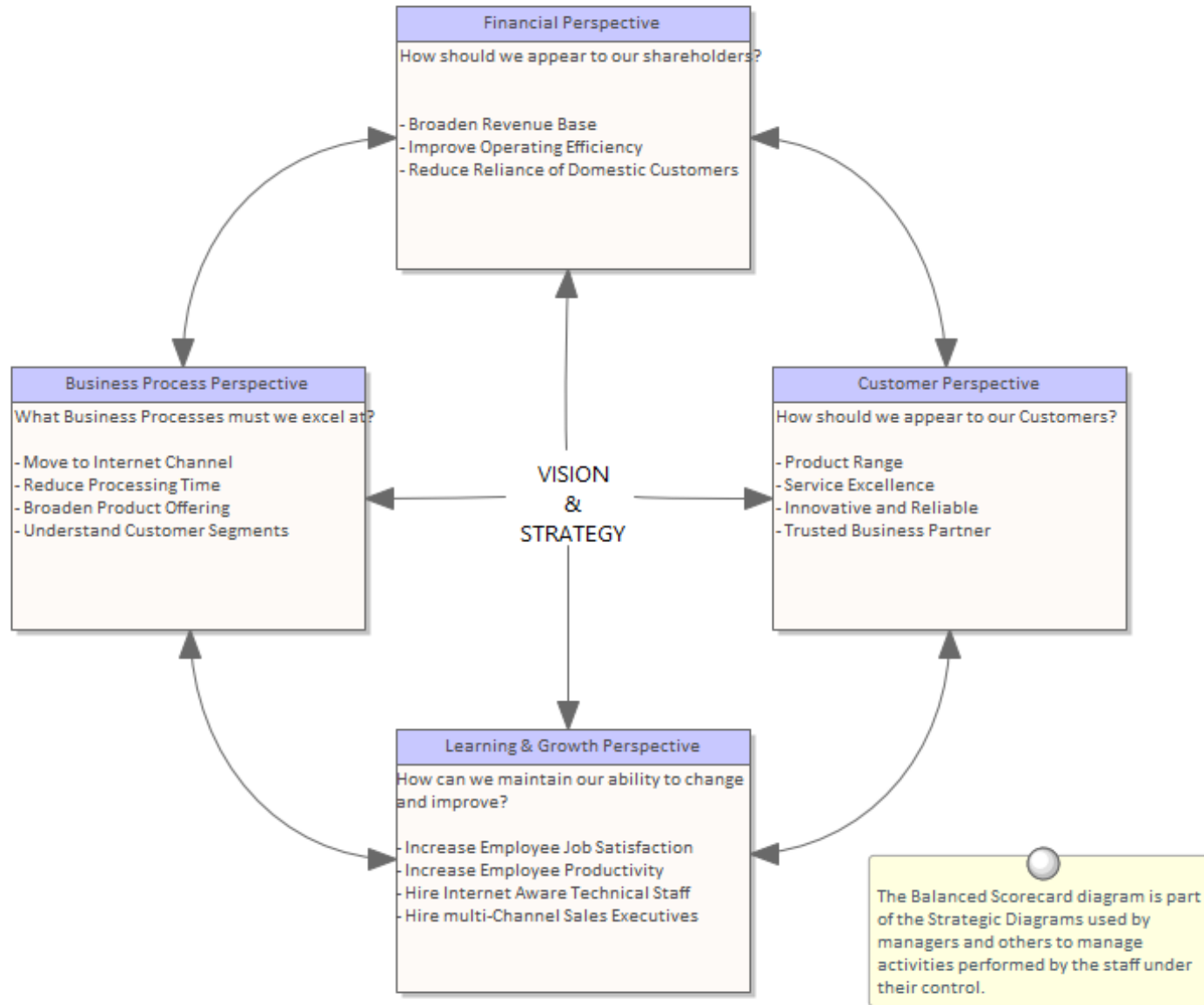
- The **Customer perspective** asks how the enterprise should appear to its customers, with measures like customer satisfaction.
- The **Financial perspective** is focused on the business value created by the enterprise, entailing measures such as shareholder value.
- The **Internal Business Processes perspective** looks at the effectiveness and efficiency of a company's internal operations, paying special attention to the primary, mission-oriented processes.
- Finally, **the Learning and Growth perspective** addresses the corporate and individual ability to change and improve, which is critical to any knowledge-intensive organisation.

ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

For each of the four perspectives the BSC proposes a **three-layered structure**:

- 1) **mission** (e.g., to become the customers' preferred supplier);
- 2) **objectives** (e.g., to provide the customers with new products);
- 3) **measures** (e.g., percentage of turnover generated by new products).

balanced scorecard Balanced Scorecard



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ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

Business Model Development

The Business Model Canvas (Osterwalder and Pigneur 2010) is a template to create high-level descriptions of new or existing business models. The Business Model Canvas consists of several parts:

- **Value proposition**: the centre of the canvas, describing what products and services an organisation has to offer to its different customers
- **Key activities**: what the organisation needs to do to provide its value propositions
- **Key resources**: the resources needed for these activities
- **Customer segments**: the typical customer groups the organisation distinguishes
- **Customer relationships**: the kind of links the organisation has with its customers
- **Channels**: how the organisation gets in touch with its customers.
- **Key partners**: others with which the organisation cooperates in delivering value to its customers
- **Cost structure**: the financial view of the means employed by the organisation
- **Revenue streams**: the way the organisation makes money from various revenue flows from its customer segments

ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

Business model canvas

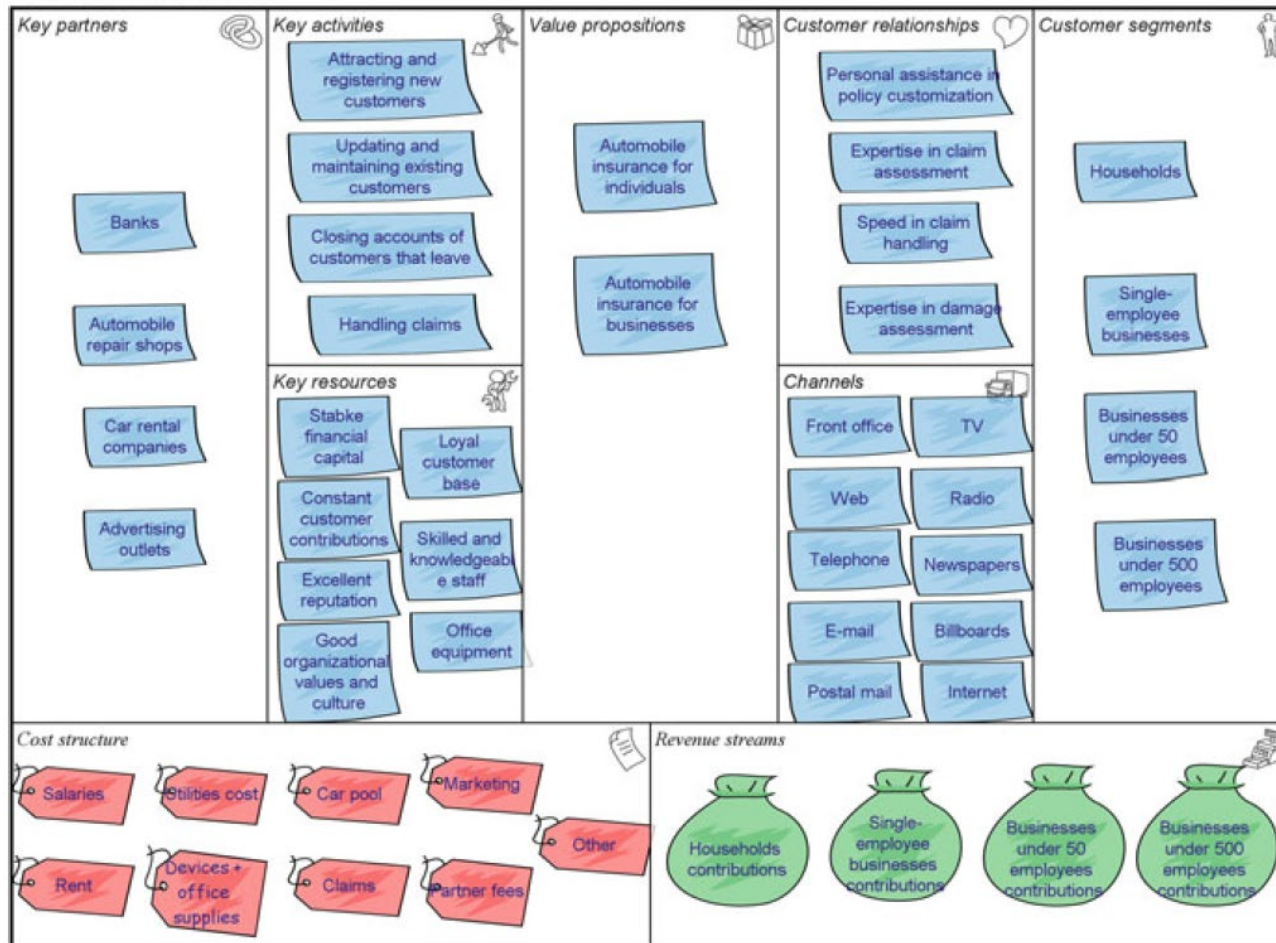


Fig. 2.2 Example business model canvas

ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

Business Architecture

A blueprint of the enterprise that provides a common understanding of the organisation and is used to align strategic objectives and tactical demands.

A business architecture provides a business-oriented abstraction of the enterprise in its ecosystem, which helps to translate strategy into action. The role of model-based support for design, analysis and decision making is becoming increasingly important in the business architecture discipline.

ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS



Fig. 2.3 The business ecosystem as represented by business architecture (Business Architecture Guild 2016)

ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

Discussion

According the below paper describe **What are the characteristics of good enterprise architecture?**

- ❖ Khayami, R. (2011). Qualitative characteristics of enterprise architecture. Procedia Computer Science, 3, 1277-1282.



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